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Call Reporting System (CRS)

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Executive Summary

In contemporary organizational environments where client communication is pivotal, monitoring and analyzing telephonic interactions has become essential for enhancing operational efficiency, service quality, and strategic planning. Conventional methods of call tracking—often manual and fragmented—pose significant challenges, including data inconsistency, time inefficiency, and limited analytical insight.

To address these limitations, the **Call Reporting System (CRS)** has been developed as a comprehensive and automated solution. This system is designed to streamline the process of collecting, storing, analyzing, and visualizing call-related data. Leveraging Microsoft Power BI for data visualization and Microsoft SQL Server for backend data management, the CRS facilitates the generation of detailed and dynamic reports that support informed decision-making.

The system offers a user-friendly interface and customizable dashboards, enabling stakeholders to monitor key metrics such as call volume, duration, agent performance, and caller trends in real time. By automating the reporting process, the CRS minimizes human error, enhances data accessibility, and improves the overall transparency of communication activities within an organization. As a robust and scalable reporting tool, the CRS represents a significant advancement in operational oversight, allowing organizations to make data-driven decisions, allocate resources effectively, and continuously improve customer interaction strategies. This system contributes to the broader objective of integrating intelligent data analytics into everyday business processes.

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Abbreviations

Abbreviation	Full Form
CRM	Customer Relationship Management
PBIX	Power BI File Extension
DAX	Data Analysis Expressions
ETL	Extract, Transform, Load
KPI	Key Performance Indicator
UI	User Interface
UX	User Experience
SQL	Structured Query Language
CSV	Comma-Separated Values
API	Application Programming Interface
BI	Business Intelligence
AI	Artificial Intelligence
ML	Machine Learning
SLA	Service Level Agreement
QA	Quality Assurance
RLS	Row Level Security
PBID	Power BI Desktop
UAT	User Acceptance Testing
DBMS	Database Management System
NPS	Net Promoter Score

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Chapter 1

Introduction

1.1 Background of Call Reporting

To assess how well customer service operations are performing, call centre reporting is crucial. In contrast to conventional phone centre reporting, it uses a broader range of indicators and channels to assess the whole customer experience. Voice-only call centres have given way to omnichannel support in modern contact centres, allowing clients to communicate with agents via phone, email, SMS, web chat, and social media [1].

In present corporate life, call centre reporting is an essential procedure, especially in the customer service sector. It includes the methodical monitoring, evaluation, and display of metrics and key performance indicators (KPIs) pertaining to call centre operations and agent effectiveness. Call centre reporting gives managers and other stakeholders important information about operational effectiveness, agent productivity, and customer satisfaction levels, which is useful for making decisions inside the organisation.

Reporting is essential for evaluating and improving several aspects of performance in the field of call centre operations [2]. With an all-encompassing perspective on call centre operations, it gives managers the ability to track call volumes, keep an eye on service levels, assess agent responsiveness, and determine the overall efficacy of operations. The evolution of call centre reporting has been driven by advancements in technology, the proliferation of digital communication channels, and the increasing emphasis on customer-centricity [3]. Traditional reporting methods, reliant on manual data collection and spreadsheet analysis, have given way to sophisticated reporting tools and analytics platforms that offer real-time insights and predictive capabilities. These technological innovations have revolutionised the way organisations approach call centre reporting, enabling them to adapt swiftly to changing customer preferences and market dynamics.

Technological improvements, the growth of digital communication channels, and the growing emphasis on customer-centricity have all contributed to the evolution of call centre reporting [4]. Sophisticated reporting tools and analytics platforms that provide real-time insights and predictive capabilities have replaced traditional reporting methods that rely on human data collecting and spreadsheet processing [5]. The way businesses handle call centre reporting has completely changed as a result of these technological advancements, which allow them to quickly adjust to shifting consumer demands and market conditions [6].

1.2 Challenges in Traditional BI Infrastructure

Business Intelligence (BI) tools have revolutionised the business landscape, with recent studies indicating a significant 26 per cent increase in sales for companies embracing advanced visualisation, dashboards, and reporting tools. Despite this, many organisations deploy these dashboards not for practical utilisation but rather for a semblance of comprehensive business understanding, highlighting a fundamental weakness of traditional BI tools. Even at their best, traditional BI tools may overlook significant issues within the vast expanse of Big Data they process. Consequently, several factors contribute to the limitations of traditional BI tools in effectively addressing the problems they are designed to tackle [7].

Call centre reporting presents multiple challenges for traditional BI infrastructure. One major drawback is the challenge of delivering real-time data, which delays insights and decisionmaking procedures that are essential for quickly resolving problems. Furthermore, Inmon and Linstedt state that call centre reporting presents multiple challenges for traditional BI infrastructure, such as the delay in delivering real-time data and the complexity of data integration due to the variety of data sources [8]. Scalability problems emerge as call centre activities grow, resulting in lower system efficiency and performance bottlenecks. Furthermore, because traditional BI systems are rigid in their approach to handling metric modifications, they frequently fail to adjust to the quickly changing reporting requirements of contact centres. As a result, pertinent insights are not reflected in a timely manner. Because traditional BI systems have a steep learning curve, access to reporting capabilities is therefore limited and mainly restricted to technical workers, which prevents non-technical stakeholders from widely adopting them [9]. Moreover, it takes a lot of time and money to maintain and update traditional BI systems, which raises operating expenses and extends the lead time for making any changes. There are extra problems in ensuring strong data security and governance, particularly when it comes to handling sensitive customer information in call centre datasets. Lastly, in dynamic call centre environments, strong predictive analytics skills are essential for proactive decision-making and resource allocation, but they may be absent from standard BI systems. Moving towards more flexible and agile BI infrastructures, like Agile BI, may be necessary to address these issues and better meet the changing requirements of call centre reporting [10].

1.3 Rationale for Agile BI Implementation

OLAP, Data Warehousing, Datamart's, Datamining, Executive Information Systems, Decision Support Systems, Neural Networks, Expert Systems, and Balanced Scorecards are just a few

of the many acronyms, technologies, and disciplines that are included in the broad field of business intelligence (BI). It might be difficult to give accurate definitions because the terminology that falls under the umbrella of business intelligence is interrelated and occasionally used interchangeably [11]. However, BI has three things in common, despite its complexity: it provides information for managing business operations, aids in decision-making, and uses a business-oriented vocabulary to communicate information. BI is an essential part of the larger management information system because it makes it easier to control corporate operations regardless of where the data is stored. The seamless running of business operations is guaranteed by this technology [12]. Operating procedures in a traditional organisational structure are susceptible to outside influences and environmental changes, such as changes in the market, new laws, or the introduction of replacement items. Operational processes require oversight, modification, and correction in response to these dynamics. Businesses run the risk of collapsing into chaos and disarray without this kind of supervision [13].

Business intelligence (BI) is seeing a radical transition in the field of call reporting, moving away from conventional engineering models that follow a step-by-step sequence of analysis, design, construction, testing, and deployment. These traditional models frequently place a premium on data and technology over communication between developers and business users. Instead of addressing basic business questions like "What business questions do we want to answer with the available data to support decision-making?" and "What are the actual needs of our users?" developers are primarily focused on data and technology. Using agile approaches is helping businesses get faster returns on their BI investments and adjust more quickly to changing business needs [14].

An agile foundation for BI implementation and administration is necessary to fully realise the promise of agile business analytics in the context of call reporting. Although there could be differences in strategy, the fundamental process is always the same. Creating a loose BI vision is called conceptualization [15]. Whiteboard meetings are frequently used for the first conversations to lay out the fundamental architecture, evaluate parts of project delivery that are realistic, and set priorities. BI funding and support decisions, active stakeholder involvement, education on agile principles, identification of critical business requirements and needs, investigation of available data sources, and planning of anticipated information delivery channels are all hallmarks of the inception stage [16].

The following one is to prioritise the most important business goals and requirements while keeping time and money limits in mind. Creating user stories for each business question that

has been identified and using a frequency vs. difficulty quadrant to establish priorities are two successful prioritisation strategies [17]. Stories in the upper right quadrant are considered high importance since they address common and difficult-to-answer business questions. On the other hand, the questions in the lower right corner, which stand for easier yet still very popular questions, might offer some promising low-hanging fruit.

1.4 Research Limitations and Problem Statement

Research on call center reporting has uncovered several significant issues with the traditional business intelligence architecture, which hinder the effective operation of modern call centers [18]. Major problems with traditional business in call centers include inflexibility in adapting to evolving demands, difficulty integrating diverse data sources, limited access to real-time analytics, and inefficiencies due to outdated tools and manual reporting. These problems are essential for properly satisfying client expectations and adapting to the industry's changing demands. Thus, the study focus revolves around the following issues:

- Conventional BI systems frequently don't have the adaptability needed to change with the demands of call centre operations. This leads to inflexible reporting systems that are difficult to alter and update instantly. When call centre requirements evolve, inefficiencies in reporting and decision-making processes result from BI systems' incapacity to adapt effectively.
- Contact centres collect a tonne of data from a variety of sources, such as social media interactions, emails, chats, and phone conversations. Significant obstacles arise when integrating and analysing data from these different sources. The inability of traditional business intelligence (BI) solutions to successfully dissolve data silos and integrate information leads to fragmented insights and inadequate analysis.
- It might be challenging to obtain real-time analytics and insights using traditional BI solutions due to latency problems. Timely access to actionable information is essential for promoting proactive decision-making and enhancing operational efficiency in a fast-paced call centre setting. The lack of access to real-time data makes it more difficult for agents and call centre managers to react quickly to shifting market conditions and client requests.
- Outdated analytics tools and manual reporting procedures are two factors that lead to inefficiencies in contact centre operations. In the absence of efficient reporting processes and user-friendly analytics dashboards, managers and agents find it difficult

to obtain pertinent data and recognise performance patterns. In addition to decreasing productivity, this inefficiency makes it more difficult for call centres to provide the best possible client experiences.

Developing an effective BI that satisfies the changing requirements of contemporary contact centres requires addressing three fundamental issues. Call centres may increase operational effectiveness, strengthen decision-making procedures, and provide better client experiences in a cutthroat market by overcoming these obstacles.

1.5 Applications of Agile BI Infrastructure in Call Center Reporting

1.5.1 Real-Time Performance Monitoring

Call centre managers may now track agent performance data in real time because of agile BI technology. Managers may monitor critical performance metrics including call resolution times, customer satisfaction ratings, and service-level agreement compliance using dynamic dashboards and customisable reports. In order to increase overall performance and customer happiness, management can take proactive measures such as identifying potential difficulties, promptly coaching agents, and implementing continuous improvement efforts, all made possible by real-time monitoring.

1.5.2 Predictive Analytics and Forecasting

Call centres can use predictive analytics models to foresee future trends and customer behaviours based on past data patterns by utilising agile BI technologies. Call centres can estimate future call volumes, identify peak times, and allocate resources appropriately by examining past call volumes, consumer questions, and agent performance indicators. In order to improve operational efficiency and resource utilisation, call centres can also use predictive analytics to identify possible bottlenecks, optimise personnel numbers, and proactively handle consumer needs.

1.5.3 Omnichannel Insights

Call centres engage with clients through a variety of communication channels in the modern omnichannel world, such as social media, emails, chats, and phone calls. The integration of data from these various channels is made easier by agile BI infrastructure, which gives call centres a thorough understanding of consumer preferences, communication styles, and channel

efficacy. Call centres may maximise customer engagement and happiness by evaluating the impact of marketing initiatives, identifying emerging trends, and optimising channel allocation plans through the analysis of omnichannel data streams.

1.5.4 Self-Service Analytics

Agile BI systems provide self-service analytics capabilities to frontline employees, such as agents and team managers. Frontline personnel can explore data, create bespoke reports, and extract actionable insights without requiring considerable IT assistance because of interactive dashboards and accessible user interfaces. Within the contact centre, self-service analytics cultivates a culture of data-driven decision-making by empowering agents to recognise patterns in performance, monitor specific metrics, and make focused adjustments to improve customer relations and operational effectiveness.

Numerous applications and advantages can be obtained by using an agile BI architecture in call centre reporting, such as self-service analytics capabilities, omnichannel insights, predictive analytics, and real-time performance monitoring. In today's cutthroat business environment, call centres may enhance resource allocation, streamline operational procedures, and provide outstanding client experiences by utilising these cutting-edge skills.

Chapter 2

Literature Review

The objective of the literature review is to explain the Agile Business Rule Development (ABRD) process and how it is used in business applications. The advantages and real-world uses of Agile principles in BI initiatives are also covered. The evaluation highlights the agile approaches' iterative development, collaborative approach, flexibility, and adaptability. The review of literature is relevant to the thesis *An Agile BI Infrastructure for Call Centre Reporting* since it provides details about agile approaches, namely the ABRD methodology, and how they may be used in the context of call centre reporting and business intelligence. It provides the foundation for understanding how call centre reporting and BI infrastructure difficulties can be addressed by applying agile concepts.

With a focus on deploying BRMS (Business Rules Management Systems), BRE (Business Rule Engines), BPEL (Business Process Execution Language), and BPM (Business Process Management) components within business applications, the Agile Business Rule Development (ABRD) methodology is an incremental and iterative approach to software development. ABRD places a strong emphasis on flexibility and cooperation between IT and business partners, in contrast to traditional waterfall models like the V-model, which frequently produce late, overbudget solutions that fall short of changing business requirements [19]. Conventional models frequently isolate business users from the development process, leading to protracted cycles for implementing changes and a lack of assurance about how well software modifications correspond with business requirements [20].

During rule discovery, analysis, and validation, active communication between rule developers, subject matter experts (SMEs), and business users is prioritised by ABRD, which follows to the Agile Alliance manifesto's guiding principles. Compact processes are used to promote effective teamwork [21]. Every ABRD iteration results in a tested set of executable rules, offering real business value and guaranteeing that business users can rely on the deployed system to precisely represent specified rules and procedures. Through active collaboration with the development team and the direct definition of business policies and norms by subject matter experts, ABRD does away with the necessity for formal specification documents [22]. The Agile Business Rule Development (ABRD) methodology is designed to deploy Business Rules Management Systems (BRMS), Business Rules Engines (BRE), Business Process Execution Language (BPEL), and Business Process Management (BPM) components within business applications. It is an incremental and iterative approach to software development. By offering a collaborative framework amongst IT business stakeholders that goes beyond the constraints of conventional waterfall software development life cycles (SDLC), ABRD meets the changing

needs of contemporary software development. Although techniques like as the waterfall or V-model are appropriate for projects that are mission-critical, in fast-paced environments where needs are constantly changing, they usually lead to overbudget, delayed solutions that fall short of the initial business objectives [23]. These conventional methods decrease agility and competitiveness by giving business users little ownership over the solution and requiring months to implement changes.

The Agile Business Rule Development (ABRD) methodology is designed to deploy Business Rules Management Systems (BRMS), Business Rules Engines (BRE), Business Process Execution Language (BPEL), and Business Process Management (BPM) components within business applications. It is an incremental and iterative approach to software development. By offering a collaborative framework amongst IT business stakeholders that goes beyond the constraints of conventional waterfall software development life cycles (SDLC), ABRD meets the changing needs of contemporary software development [24]. Although techniques like as the waterfall or V-model are appropriate for projects that are mission-critical, in fast-paced environments where needs are constantly changing, they usually lead to overbudget, delayed solutions that fall short of the initial business objectives. These conventional methods decrease agility and competitiveness by giving business users little ownership over the solution and requiring months to implement changes [25].

The Agile Business Rule Development (ABRD) approach is based on the Agile Alliance philosophy, which emphasises client involvement, tangible outcomes, and teamwork.

2.1 Individuals and Relationships over Procedures and Tools:

Throughout the phases of rule discovery, analysis, and validation, ABRD encourages open communication between rule creators, subject matter experts (SMEs), and business users. ABRD ensures that the development process is adaptable and responsive to changing requirements by placing a higher priority on direct interaction than inflexible procedures [26].

2.2 Functional Software as compared to Detailed Documentation:

Providing tested rule sets for every iteration is a core principle of ABRD. ABRD places more emphasis on the development of concrete, workable solutions than it does on detailed documentation alone [27]. By using this method, stakeholders are reassured that the implemented system accurately reflects established guidelines and procedures.

2.3 Client Collaboration vs. Contract Negotiation:

SMEs are actively included in the development process by ABRD, giving them direct control over the definition of corporate policies and procedures. This cooperative method guarantees that the final system successfully satisfies business needs and does away with the necessity for formal specification documentation. ABRD promotes cooperation between the development team and business stakeholders, which helps to create a common understanding of the objectives and specifications of the project [28].

The ABRD process is implemented in a sequence of iterative cycles, with each cycle being customised for certain development activities:

Cycle 1: Harvesting: During this first stage, the project team primarily concentrates on comprehending business entities and recording implementation guidelines. Rule analysis and discovery are two business modelling tasks that aid in locating possible rule patterns and decision points in the business process.

Cycle 2: Prototyping: During this cycle, the project is organised and guidelines are set. In order to make sure that the changing rule set is in line with business goals, early rule authoring enables the detection of problems and feedback loops with business stakeholders.

Cycle 3: Building: The building phase prioritises rule creation, validation, and the execution of unit test cases using actual data in accordance with Test-Driven Development (TDD) principles. Project acceptability and progress are governed by executable rules, which come first.

Cycle 4: Integrating: In order to verify system functionality, this phase focuses on implementing the rule set for end-to-end testing, integrating the decision service with the domain data model, and running test scenarios using actual data.

Cycle 5: Enhancing: The last cycle involves authoring, validating, and deploying the rule set in order to finish and maintain it. In order to maintain the quality and completeness of the rule set over time, business stakeholders actively participate in rule refining.

Through the adoption of these guiding principles and iterative cycles, the ABRD methodology aids organizations in integrating business intelligence into call center operations, thus shortening development cycles, boosting stakeholder confidence in software alignment with business goals, and facilitating swift adaptation to changing business requirements [29].

2.4 Application of Agile Principles in BI Projects

The dynamic nature of business requirements and the need for quick insights and decisionmaking have led to a substantial increase in the application of Agile principles in business intelligence (BI) initiatives in recent years. When used in BI projects, agile approaches which are defined by flexible, collaborative, and iterative development offer several benefits. Let's discuss the possible advantages and practical applications of Agile concepts in business intelligence projects [30].

2.5 Iterative development

Agile BI projects have a strong emphasis on dividing the development process into more manageable sprints or smaller iterations. Usually lasting two to four weeks, each sprint produces a working component of the BI solution. This methodology facilitates gradual enhancements and modifications in response to input from stakeholders [31].

Table 1 : Table of Sprints

SPRINT	OBJECTIVES
1	Data extraction and cleansing
2	Data modelling and transformation
3	Dashboard design and visualisation
4	User acceptance testing and refinement

2.6 Collaborative Approach

Agile encourages cooperation between business stakeholders, data analysts, developers, and testers in cross-functional teams. Regular meetings, like daily stand-ups, sprint planning sessions, and sprint review meetings, help to guarantee alignment between technical implementation and business requirements and to improve communication [32].

Table 2: Table of Meetings

MEETING	PURPOSE
DAILY STAND-UP	Share progress, discuss challenges, plan for the day
SPRINT PLANNING	Define goals and tasks for the upcoming sprint
SPRINT REVIEW	Demo completed work to stakeholders for feedback
RETROSPECTIVE	Reflect on the sprint and identify areas for improvement

2.7 Flexibility and Adaptability

Agile approaches concentrate on adapting to change than they do on following through to a set plan. This entails adapting BI initiatives to changing data sources, analytical requirements, and business requirements. Agile teams embrace innovation and modify the priority of their projects accordingly.

Stakeholders ask for more KPIs to be added to the dashboard during a sprint review. Reordering tasks for the upcoming sprint allows the Agile team to adjust to the changing requirements. The benefits of implementing Agile concepts in BI initiatives are numerous. Frequent delivery and ongoing engagement increase stakeholder satisfaction by giving them access to observable progress and feedback channels [33]. Agile approaches also speed up the delivery of BI solutions, reducing the time to insight and enabling businesses to act quickly on data-driven choices. By doing frequent testing and validation across sprints, problems are found and fixed early on, guaranteeing improved data quality and accuracy. Additionally, agile BI projects are more flexible, and able to adapt to changing market conditions, business priorities, and data sources, maintaining the solution's usefulness and relevance over time. Agile development methods, in their most basic form, encourage iterative development, cooperation, flexibility, and adaptability, leading to BI solutions that are sensitive to the changing demands of enterprises. By following Agile techniques and implementing them with diligence, organisations can effectively utilise their data assets to guide well-informed decision-making processes [34].

2.8 Data Vault Modeling and STAR Schema

The Data Warehouse (DW) is a foundation in the field of call reporting, showing important traits such as subject-orientedness, nonvolatility, integration, and time variation, all of which

support well-informed management decisions. The Extract, Transform, Load (ETL) process is essential to most business intelligence initiatives, and DW systems excel at it. They carefully manage the extraction, transformation, and loading of data from transactional systems into the warehouse [35]. Within the DW, data marts use dimensional schemas to simplify data access and analysis, in contrast to transactional data sources that are organised in relational schemas. One of the main pillars of the DW framework is metadata, which includes definitions and guidelines for creating data and improves data understanding and governance.

Two well-known models stand out in the complex field of DW architecture: Ralph Kimball's bus architecture and Bill Inmon's Corporate Information Factory (CIF), both of which offer unique ways to data integration and storage [36]. The second-generation project, known as DW 2.0, addresses the changing requirements of contemporary data ecosystems by providing improved support for temporal data changes. The requirements of an integrated, flexible, auditable data storage are addressed by modelling methodologies like Anchor Modelling and Data Vault Modelling, with Data Vault's Hub, Link, and Satellite entities providing reliable solutions for data integrity and maintenance. The architecture of Data Vault requires complementary data mart layers leveraging star schemas for intensive query and reporting functions, ensuring rapid data access for call reporting and analysis, even though it shines in data integration and protection [37].

The star schema is a critical data modelling technique used in call reporting that is designed to enhance query performance and streamline data processing. This schema format is distinguished by a core "fact" table encircled by "dimension" tables, which, when graphically represented, resemble a star [38]. Quantitative metrics or measurements pertaining to call activities, such as call length, call volume, or call results, are usually included in the fact table. However, dimension tables hold descriptive elements like caller information, call types, timestamps, and geographic data that give context to the measurements. The call reporting scenario utilises the fact table as the central repository for crucial metrics connected to calls. Call duration, call volume, call resolution times, and other pertinent performance indicators are a few examples of the data items that could be included. A particular call event or transaction is associated with each record in the fact table. Dimension tables around the fact table, offering more detail and context for the call data. Information regarding callers (demographics, customer segments, or account kinds), call channels (phone, chat, or email), time (time stamps, dates, and time periods), and place (geographical areas or call centre branches) can all be included in these dimensions [39].

The denormalized structure and simplicity of the star schema reduce the amount of database joins needed to obtain pertinent data for analysis, making query execution more efficient. The star schema is simple to explore and query for analysts and decision-makers to get information on call performance, trends, and patterns [40].

Comparing the Data Vault and STAR Schema approaches is crucial to understanding their distinct applicability and advantages in the context of call centre reporting. With dimension tables surrounding a central fact table, the STAR Schema is known for its simplicity. This simple structure is ideal for simple reporting requirements and allows for fast query performance. But because of its simplicity, the STAR Schema might not be as flexible or scalable to handle intricate data connection scenarios or changing business needs. The STAR Schema may have issues with data redundancy and schema changes as the number of dimensions and metrics rises, which makes it less appropriate for dynamic data settings [41].

The Data Vault paradigm, on the other hand, provides a more complex yet scalable method of data modelling. The Data Vault offers a strong architecture for handling massive amounts of data and adapting to changes in data sources or business procedures over time with its Hub, Link, and Satellite tables. The Data Vault is perfect for dynamic and changing data landscapes because of its modular architecture and incremental loading mechanisms, which allow for small modifications without upsetting established structures. The Data Vault is a better option for organisations looking for a future-proof call centre reporting solution because of its flexibility and scalability, even though it requires more work to develop and maintain than the STAR Schema [42].

The STAR Schema is weaker than the Data Vault in terms of auditability and data integrity. The Data Vault guarantees complete data continuity and makes compliance with legal requirements and data governance standards easier with built-in tools for data traceability and audit trails. The Data Vault reduces the possibility of data anomalies and inconsistencies that may occur in complicated reporting environments by strengthening data quality and reliability through the enforcement of stringent business rules and validation tests. However, features like query optimisation and efficiency are quite important when choosing. With quicker response times and less complicated query execution plans, the STAR Schema is designed with query performance and real-time reporting in mind. In contrast, advanced indexing techniques and query optimisation procedures could be needed for the Data Vault to operate at peak efficiency, particularly in large-scale installations with intricate reporting scenarios [42]. There are publications that deal with using the STAR and Data Vault concepts in the contact centre

industry. Kimball and Ross (2011) conducted research named "The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modelling" which delves into the application of STAR schemas in data warehousing [43]. The study also examines the usefulness and efficiency of STAR schemas in call centre analytics and reporting. For call centre reporting, natural language processing framework weighing variables like simplicity, flexibility, data integrity, and performance must carefully assess their unique needs and priorities and decide between the Data Vault and STAR Schema [44]. In the end, the organisation's specific needs and goals for using call centre data to support strategic insights and decision-making will determine which data modelling technique is best.

2.9 Role of Python in Business Intelligence

When it comes to call centre reporting, data visualisation becomes a crucial instrument for interpreting intricate datasets and gleaning valuable information. The proverb "a picture is worth a thousand words" perfectly captures the core of data visualisation, since graphical representations help humans understand information more efficiently than rows and columns of numerical data. When information is displayed visually, people can identify trends and ask relevant questions with more ease.

Businesses can use data visualisation for a variety of reasons, including dynamic interpretation and analysis of important metrics and trends. Organisations can have a better understanding of their operational procedures and consumer interactions by converting corporate data into interactive graphs and visually appealing dashboards. Businesses may discover areas for improvement, maximise operational efficiency, and seize new possibilities by visualising patterns, trends, and correlations. With the daily generation of huge quantities of data in the modern corporate landscape, efficient data visualisation techniques are essential for guiding strategic objectives and helping to make well-informed decisions. Through process analysis and data visualisation, firms may foresee trends, find new business possibilities, and gain a better understanding of client behaviour. Additionally, data visualisation promotes coordination and cooperation throughout many organisational divisions, giving stakeholders a common understanding of critical performance metrics.

With the development of data science technology, data visualisation has the potential to become a scientific field and go beyond its previous status as an art form. Data visualisation will become more and more important in a variety of fields, such as network theory, social science research, the Internet of Things, and more, as businesses continue to use the power of multidimensional visualisations and advanced analytics. When it comes to helping businesses make decisions,

data visualisation is unmatched in its benefits. It makes it possible for them to react swiftly to shifting market conditions, make complicated datasets easier to understand, spot new trends, and encourage teamwork. Businesses may better understand their operations, allocate resources more efficiently, and spur innovation in the quickly changing digital ecosystem by utilising data visualisation approaches.

Python provides an extensive collection of modules and frameworks designed for effective data processing, analysis, and visualisation in the call reporting sector. The most advanced is Pandas, a robust toolkit for data manipulation that provides data structures like DataFrames and Series, making tabular data like call logs and customer information easier to organise and analyse. NumPy is an essential library for numerical computing that works well with Pandas. It is well-known for supporting big, multi-dimensional arrays and matrices, and its broadcasting and array operations allow for quick computation and call data handling. By providing modules for optimisation, integration, linear algebra, and statistics, SciPy expands the capabilities of NumPy and allows for sophisticated statistical analysis and modelling on call data. In order to create static, dynamic, and high-quality visualisations that reveal patterns and trends in call performance and customer interactions, Matplotlib and Seaborn become essential visualisation tools. Furthermore, Scikit-learn makes predictive analytics jobs on call data easier by providing a wide range of machine learning techniques for classification, regression, clustering, and model evaluation. Through Python's rich ecosystem of data processing tools, businesses can extract actionable insights, optimise call centre operations, and enhance customer experiences. TensorFlow and PyTorch excel as deep learning frameworks for advanced analytics endeavours, facilitating tasks like sentiment analysis of customer calls or speech recognition for call transcriptions.

Python scripts are an essential tool for call reporting in the field of Business Intelligence (BI) infrastructure development [45]. They are useful at every level of the process, from data extraction to visualisation. First of all, Python scripts make it easier to collect and combine data from a variety of sources, including external APIs, CRM systems, and call centre databases. These scripts provide consistency and correctness in the data pipeline by harmonising different datasets into a single format for analysis by utilising libraries such as Pandas and SQLAlchemy. Python scripts are essential for cleaning and preparing raw call data after it has been gathered. These scripts ensure that the data is clean and ready for analysis by handling tasks like missing value imputation, data normalisation, and outlier detection with bespoke functions and libraries like Pandas and NumPy. Additionally, call data can be improved and transformed with Python

scripts, resulting in the extraction of valuable insights. These scripts offer an all-encompassing perspective of call centre metrics through the computation of key performance indicators (KPIs) and the contextual enrichment of datasets. In order to find patterns and trends in call data, Python scripts also make complex data analysis and modelling work easier. These scripts do qualitative analysis, regression analysis, and predictive modelling by utilising libraries such as Scikit-learn, Statsmodels, or TensorFlow. This enables organisations to make data-driven decisions. Python scripts also make it possible to automate the creation of dashboards and reports for stakeholders. These scripts produce interactive charts and visualisations using libraries such as Matplotlib, Seaborn, or Plotly, which improve comprehension of call centre performance indicators. Python scripts can also be used to track call centre metrics in real time and send out alerts when performance deviates from normal or strange trends emerge. These scripts allow organisations to react quickly to new problems by using anomaly detection algorithms and statistical process control (SPC) charts.

Furthermore, Python scripts interact with well-known BI tools and platforms like Tableau, Power BI, and QlikView with ease. These scripts exploit Python's power to extract, transform, and load (ETL) data into BI systems through APIs or SDKs, enabling users to make the most of Python's capabilities inside their current BI workflows. Python scripts are flexible and adaptable technologies that are useful when developing business intelligence (BI) infrastructure for call reporting. Through the thorough analysis of call data, these scripts enable organisations to extract actionable insights, optimise operations, and improve decision-making processes. They do this by providing everything from data extraction and pre-treatment to advanced analytics and visualisation.

Chapter 3

Material and Methodology

3.1 Dataset details

The "Citizen Service Request CSR Call Centre Calls" dataset contains a huge amount of data about Cincinnati, Ohio call centre operations and service requests. Every feature in the dataset has a specific role and adds to the comprehension of how the call centre operates.

The `case_id`, a distinct number provided to every service request case that enables easy tracking and referencing within the system, is the central component of the dataset. In addition to this identification, the `service_name` and `service_subtype` elements provide more detailed information about the nature of each service request by grouping them into more general categories and particular subtypes, respectively. By including extra information provided by the caller, the description field deepens the understanding by helping to put the service requirements in context.

Fields like `case_owner` and `case_owner_group`, which identify the designated agent and their team, show how duties and responsibilities are distributed within the call centre. The efficient distribution of effort and utilisation of skills is made possible by this hierarchical design. Fields that record the start and finish timestamps of service requests, such as `case_creation_date` and `case_closed_date`, are used to record their temporal elements. In the meantime, the Service Level Agreement term is defined by the `sla_days` parameter, which also guarantees that service standards are followed and establishes expectations for prompt response. Fields such as `zipcode`, `district`, `ward`, and `community_council_neighborhood` provide geographic context; this helps to target service delivery efforts and provides insights into the spatial distribution of service requests. Furthermore, the latitude and longitude coordinates allow for accurate geographic analysis and visualisation of the distribution of service requests. The location feature helps identify the precise origin of each service request by providing comprehensive address information.

In addition, the dataset has a number of metadata fields that offer further context and documentation pertaining to the service request lifecycle, such as `closed_when_created`, `case_record_type`, `case_origin`, `case_closed_reason`, and `case_closed_date_time`. In addition, parameters like `contact_type`, `contact_subtype`, and `contact_date_time` provide information on the channels that callers use to communicate, which makes it easier to classify and monitor communication timelines. Every element in the dataset is essential to understanding and managing the operations of the call centre, guaranteeing effective customer satisfaction and service delivery.

3.2 Methodology

The process of creating an Agile Business Intelligence (BI) infrastructure that is optimised for call centre reporting is complex and requires comprehensive evaluation of a number of approaches and technologies. I understand how important it is to use Data Vault and STAR Schema Modelling concepts in BI infrastructure. Scalability, flexibility, and adaptation to modifications in data sources and business rules are given top priority in Data Vault's structured methodology. However, STAR Schema Modelling offers an enhanced schema architecture that guarantees faster data retrieval and analysis for effective reporting and querying. In order to address the specific needs of call centre reporting, I will precisely develop Data Vault structures and STAR Schema tables before initiating the implementation process. To perform this, appropriate information about entities, relationships, and attributes must be identified and organised to enable effective data storage and retrieval. Furthermore, Python scripts will be utilised to automate the loading procedure, guaranteeing precision and uniformity while expediting the integration of data from various sources.

My methodology focuses a high priority on integrating Agile methods at every stage of the development process. I prioritise ongoing feedback loops, iterative development, and stakeholder and development team communication by implementing Agile approaches like Scrum or Kanban. I can implement small, iterative changes and enhancements based on realtime input due to this iterative strategy, which eventually produces BI infrastructure that closely complies with user needs and corporate goals. Moreover, I understand how important it is to demonstrate the way our infrastructure has been put to work. As such, I will demonstrate how Power BI is integrated as the front end for visualisation. Call centre managers will be able to extract useful insights from the data due to Power BI's comprehensive reporting features and easy-to-use interface. They may effectively promote operational efficiency and make well informed judgements according to this empowerment.

3.3 State-of-the-art Data Vault

Our approach combines the reliable principles of Data Vault and STAR Schema Modelling with Agile approaches and contemporary visualisation technologies to produce a complete solution specifically designed to meet the changing demands of call centre reporting. Our mission is to transform call centre data analysis and use it for organisational success through rigorous planning, iterative development, and a dedication to generating real business value.

The Data Vault methodology provides a comprehensive strategy for all Data Warehouse Projects, extending beyond modelling methodologies. Business intelligence (BI) initiatives have historically used the waterfall approach, which is defined by a step-by-step procedure that includes gathering requirements up front, designing a comprehensive data model, and developing the visualisation layer in turn. Unfortunately, this method frequently results in lengthy project schedules and makes it challenging to adjust to evolving business requirements. Agile approaches have gained popularity as a solution to these problems because they emphasise iterative development and adaptability. However, in the field of data analytics, agile might not be sufficient to handle the intricacies of BI or data warehouse projects on its own. In his book "Building Scalable Data Warehouse with Data Vault 2.0," Daniel Linstedt presents a methodology that combines established proven disciplines and methodologies with agile principles. By tackling issues like iterating over data modelling, reducing refactoring, and creating ETL (Extract, Transform, Load) routines that can quickly react to changes in business logic or new data additions, this strategy seeks to improve agility in data analytics initiatives. The Data Vault methodology offers a more iterative approach to BI projects by incorporating these techniques, which helps teams deliver results more quickly.

Neglecting common opinion, Data Vault (DV) is not merely a modelling tool; rather, it is a complete methodology for data warehouse initiatives. This methodology combines aspects of data vault modelling, agile approaches, BEAM requirements gathering, CMMI, TQM, Six Sigma, and TQM with the goal of improving the effectiveness and calibre of business intelligence (BI) initiatives. DV, which is sometimes compared to a "guided missile approach," focuses the same importance on project execution precision and flexibility.

Agile methodologies have been integrated into the DV methodology to estimate project deadlines and assess task complexity for common Data Warehouse components. It provides an organised and iterative method for handling technical deliverables in the BI domain at a detailed level, especially when responding to requests for new or changing features. For managing standard tasks like adding data attributes, integrating new data sources, tracking historical data, and changing source structures during both the ETL and Modelling stages, involves well-defined, repeatable, and agile-driven procedures.

The Data Vault 2.0 Approach offers a thorough approach with several benefits. Assuming worst-case scenarios, it first proactively addresses potential complications in data modelling relationships. It minimises extra effort and ensures adaptability by supporting N:M relationships between business items, which lessens the requirement for significant adjustments

when relationship degrees change. Second, the strategy is designed to provide extensive tracking of data properties and sourcing operations throughout time in the past. Data integrity is ensured, and trend analysis is made easier by retaining past data changes through the use of satellites, which function similarly to SCD Type 2. Thirdly, an established structure for improving historical tracking performance within the Vault is provided by Data Vault 2.0. Bridge tables and Point-in-Time (PiT) are two examples of structures and design ideas included in this framework that may be easily incorporated into the modelling process without requiring a great deal of preplanning. The method also places a strong emphasis on the logical division of areas holding modified and raw data. While the Business Vault offers a controlled environment for power users needing access to refined data, the Raw Data Vault provides the basis for auditable data sourcing from several systems.

Moreover, Data Vault encourages the division of business rules into two distinct categories when integrating data. This division reduces duplication in data processing operations and improves the reusability of data across several applications. Furthermore, the methodology facilitates incremental extension without compromising the integrity of previous data, which supports agile development practices. Since historical tracking is kept separate from the dimensional model, it can be easily expanded upon and changed throughout agile iterations. Furthermore, Data Vault promotes using hash keys for business keys to maximise loading parallelization and minimise lookup dependencies, improving loading performance and efficiency. Additionally, the Raw Data Vault is made to be completely auditable, guaranteeing accountability and transparency in data processing operations. Data Vault simplifies the conversion from unprocessed data to refined forms like OLAP cubes and Star Schema. It is an excellent choice for combining diverse data sources with different business objectives because of its flexible and iterative approach. It also naturally aligns with agile development processes and guarantees effective and efficient data management procedures.

3.4 Conversion into Data Vault Model

In this thesis, I delve into the process of converting raw call center data into a structured Data Vault model. By analyzing the relationships between various data columns and identifying core business entities, I establish a robust framework for storing and managing call center data effectively. The model comprises Hubs, Links, and Satellites, each serving distinct roles in encapsulating essential business entities, their relationships, and supplementary descriptive

information. Through this breakdown, I present a comprehensive approach to transforming complex call center data into a streamlined and scalable architecture.

Hubs:

AGENTDISPID: Represents the unique identifier for agents.

CALLACTIONID: Represents the unique identifier for call actions.

CALLTYPEID: Represents the unique identifier for call types.

SERVICE_ID: Represents the unique identifier for services.

STATION: Represents the unique identifier for stations.

WORKGROUP_ID: Represents the unique identifier for workgroups.

Links:

CALLID: Connects various components of a call (e.g., agent, call action, call type).

SERVICE_ID - WORKGROUP_ID: Represents the relationship between services and workgroups.

Satellites:

CALLSTARTDT: Satellite for capturing additional information about call start date and time.

CALLACTIONREASONID: Satellite for capturing additional information about call action reasons.

CONNCLEARDT: Satellite for capturing additional information about connection clear date and time.

DNIS: Satellite for capturing additional information about DNIS (Dialed Number Identification Service).

QUEUEENDDT: Satellite for capturing additional information about queue end date and time.

SEQNUM: Satellite for capturing additional information about sequence numbers.

WRAPENDDT: Satellite for capturing additional information about wrap end date and time.

QUEUESTARTDT: Satellite for capturing additional information about queue start date and time.

PREVIEWENDDT: Satellite for capturing additional information about preview end date and time.

PREVIEWSTARTDT: Satellite for capturing additional information about preview start date and time.

ANSWERDT: Satellite for capturing additional information about answer date and time.

ANSWER_SPEED_SECS: Satellite for capturing additional information about answer speed in seconds.

TALK_TIME_SECS: Satellite for capturing additional information about talk time in seconds.

WRAP_TIME_SECS: Satellite for capturing additional information about wrap time in seconds.

SERVICE_LEVEL: Satellite for capturing additional information about service level.

ABANDONED: Satellite for capturing additional information about whether the call was abandoned.

CALL_TYPE: Satellite for capturing additional information about call types.

UNIQUE_ID: Satellite for capturing additional information about unique identifiers.

ANSWERED: Satellite for capturing additional information about whether the call was answered.

This breakdown organizes the data columns into Hubs, Links, and Satellites, which form the foundation of a Data Vault model as shown in Figure 1. The Hubs represent core business entities, the Links capture relationships between entities, and the Satellites store additional descriptive information about the entities and their relationships.

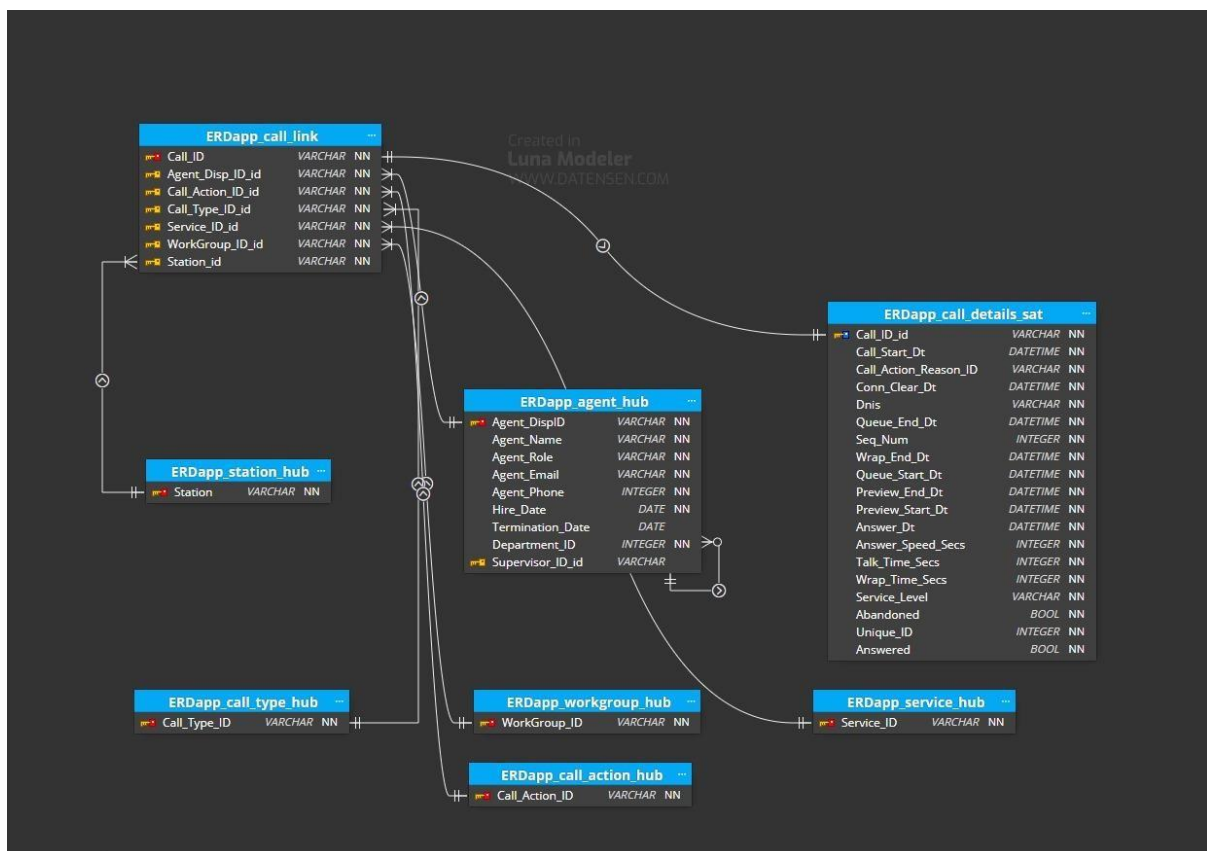


Figure 1 Data vault schema of the proposed work

3.5 State-of-the-art STAR Schema

A star schema is an organised approach to arranging data in a database such that it can be easily understood and analysed. This approach may be used for a wide range of data management systems, such as databases, data marts, and warehouses, and it offers enhanced querying capabilities to handle large datasets. Star schemas, first introduced by Ralph Kimball in the 1990s, minimise redundancy in business definitions, making them effective for data storage, historical maintenance, and upgrades. The efficiency of aggregating and filtering data inside the data warehouse is improved by this decrease in duplication. A star schema divides complicated business data into facts (such as transaction details like amounts and quantities) and dimensions (like time, and product).

A single fact table, which is at the centre of the star schema architecture and contains crucial business data like transaction details, is central to the design. This fact table links to other dimension tables that show different elements, such as time or product. As a result, by merging these fact and dimension tables through slicing and dicing processes, users can analyse data from various angles. This adaptability enables flexible data analysis and exploration, meeting a range of database environment query requirements. Star schemas are essential for improving the effectiveness of reporting and data querying, especially in call centre settings. Repetitive columns are added to some dimension tables when data is denormalized, making data retrieval and analysis quicker and simpler. This trade-off avoids computationally costly join operations by surrendering a little amount of redundancy in the data model to speed up query execution. Star schemas preserve denormalized dimension tables and normalised fact tables in the context of call reporting. This indicates that although information in dimension tables could be repetitive, data in fact tables stays unique. A structure like this makes it easier for end users to obtain data and facilitates the effective retrieval of information relevant to call centre performance. Star schemas are very beneficial to call reporting because of the simplicity and convenience with which they are implemented. They make it easier for end users to find and evaluate the data they need, which improves call centre operations' general efficiency. Furthermore, since they don't rely as much on join operations, they can handle basic queries, which is important in the fast-paced world of call reporting. Star schemas also easily interface with Online Analytical Processing (OLAP) models, which increases their usefulness for call centre data analysis. In comparison to normalised data models like snowflake schemas, their design prioritises enhanced query efficiency, and they strive to reduce the requirement for computationally demanding joins. This query performance optimisation is especially helpful

for call reporting, where decision-making and performance evaluation depend on having access to correct data.

3.6 The Agile Framework and Its Application

The Agile Framework stands as a dynamic methodology revolutionizing software development and project management, particularly addressing the rigidity of conventional approaches. Agile emphasizes flexibility, teamwork, and iterative development, contrasting with traditional methods' strict planning and documentation requirements. This shift promotes continual improvement and customer satisfaction, making it particularly relevant in industries marked by constant change, such as customer service centers' call reporting.

At the core of Agile lies the Agile Manifesto, a foundational document outlining its principles and ideals. Agile promotes a departure from rigid, linear approaches towards more flexible and responsive methodologies. In the realm of call reporting within customer service departments, where regulations are in a constant state of flux, Agile's embrace of change and uncertainty presents opportunities for progress rather than obstacles.

Agile fosters cooperation and cross-functional teams, leveraging diverse talents and perspectives to tackle complex challenges effectively. In the context of call reporting, this collaborative approach ensures that reporting solutions are not only technically robust but also aligned with the needs and expectations of stakeholders.

One of Agile's distinctive features is its iterative approach to development, characterized by time-limited iterations or sprints. This iterative process enables teams to deliver functional software early and frequently, facilitating feedback from stakeholders and adjustments as necessary. In the realm of call reporting, this iterative method allows for the rapid integration of new metrics and reporting features in response to evolving business needs.

Agile encompasses various methodologies, including Lean, Kanban, and Scrum, each offering its own set of guidelines and best practices suited to specific project situations. For instance, Scrum structures development processes through defined roles, events, and artifacts, while Kanban focuses on workflow visualization and work-in-progress limitation. Lean, inspired by lean manufacturing principles, aims to maximize value delivery while minimizing waste. Projects within customer service centers, particularly those involving call reporting, benefit greatly from Agile's flexible and adaptive approach to project management and software development. By embracing Agile concepts and practices, organizations can enhance their ability to provide high-quality solutions that meet customer needs and drive business value.

Case Studies:

- Highlight success stories of customer service centers implementing Agile methodologies for call reporting, showcasing how Agile practices led to tangible improvements in reporting accuracy, efficiency, and responsiveness to changing requirements.

Agile Practices for Call Reporting:

- Agile practices tailored to call reporting, such as sprint planning for prioritizing reporting features, daily stand-ups for ensuring collaboration and transparency, and retrospective meetings for continuous improvement.

Integration with BI Tools:

- Agile methodologies can be integrated with BI tools commonly used in call reporting, such as dashboarding platforms or analytics software, to enable iterative development and continuous enhancement of reporting capabilities.

Scalability and Adaptability:

- Agile frameworks like Scrum or Kanban can be scaled to accommodate the evolving needs and complexities of call reporting projects within customer service centers, fostering agility and resilience in the face of change.

3.7 Implementation of data vault and STAR schema in Python

Data extraction, transformation, and loading into the appropriate structures are accomplished through a detailed set of procedures in the Python data filling processes for Data Vault and Star Schema. Developers can create scripts and programmes that automate these crucial operations by utilising Python's powerful capabilities and broad ecosystem of modules and tools. The first step is to connect to various data sources, which is made easier by Python packages such as NumPy, Pandas, and SQLAlchemy. These libraries enable programmers to communicate with a variety of data sources, including flat files, relational databases, and APIs. Through the utilisation of these features, information may be effectively extracted in an organised fashion from various sources, providing a foundation for further processing and loading procedures.

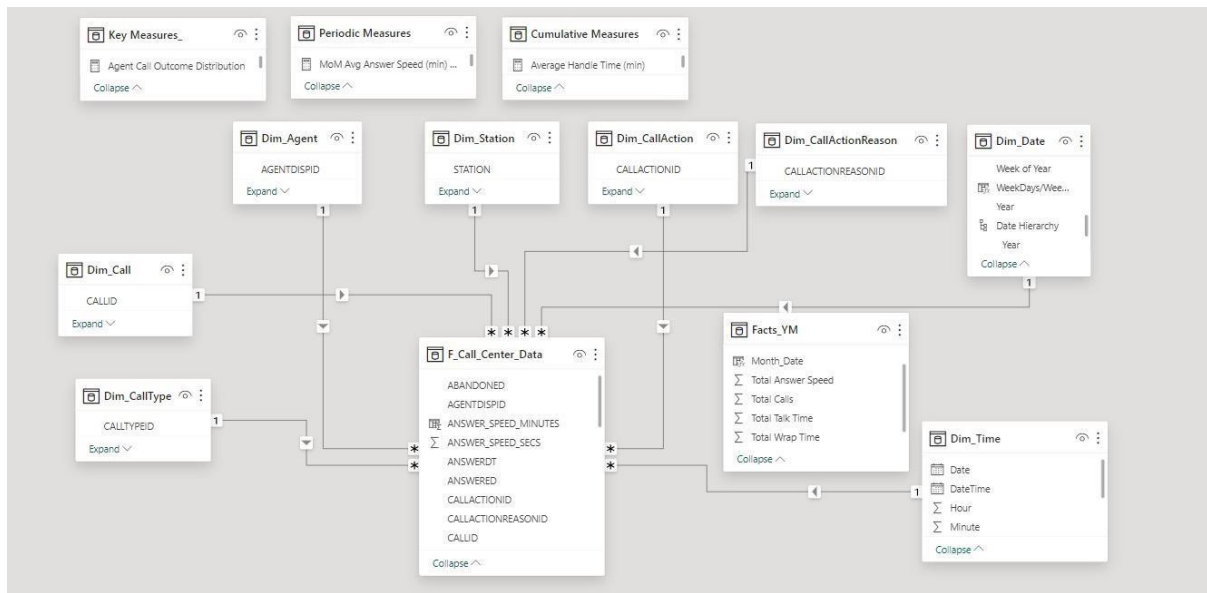


Figure 2 Star schema of the proposed work

To validate consistency and compatibility, data must go through a critical transformation procedure before being loaded into the Data Vault tables. In order to maintain data quality requirements, this transformation typically involves actions like data cleansing, normalisation, and the application of business rules. Python proves to be a potent instrument for carrying out these changes efficiently, as programmers create scripts that utilise libraries such as Pandas to achieve smooth data transformation and manipulation. Python scripts may effectively perform activities like data cleansing, restructuring, and validation by utilising Pandas' features, which guarantees that the data meets the requirements of the Data Vault model. Loading the converted data into the Data Vault tables is the next step after the transformation process. Many libraries, such as SQLAlchemy or PyODBC, are available for Python, allowing for easy database interface and the execution of SQL queries to put the changed data into the appropriate tables inside the Data Vault architecture. Strong error handling techniques must be included in order to handle exceptions efficiently and keep the loaded data intact at all times.

Similar transformations are required for the retrieved data to match the denormalized structure of the Star Schema. The data can be aggregated and denormalized using Python routines, readying it for loading into the Star Schema tables. The transformation process might additionally include data enrichment and validation stages to improve the quality and usability of the data in the Star Schema. There are general techniques that apply to both Data Vault and Star Schema implementation processes in addition to the specific ones. These include adding data quality checks to guarantee the accuracy and completeness of the data that has been loaded, coming up with incremental loading plans to efficiently handle updates, and automating the

ETL procedures with frameworks like Apache Airflow for more efficient scheduling and monitoring. Robust monitoring and logging systems are essential for carrying out data filling procedures, giving developers information about the efficiency of ETL operations, identifying any problems and assisting with error troubleshooting. Developers may optimise the generation of Data Vault and Star Schema structures and enable organisations to derive important insights from their data resources by carefully following these procedures and utilising Python's capabilities.

Chapter 4

Results and

Discussion

4.1 Overview of Dashboard

Following dashboard were designed to analyse the performance of call centre operations via intelligent dashboard system.

4.1.1 Call centre overview

Provides real-time insights into call center performance, utilizing datasets including CALLSTARTDT for call initiation, ANSWERDT for call response times, and SERVICE_LEVEL for service quality. With detailed metrics like TALK_TIME_SECS and WRAP_TIME_SECS, stakeholders can analyze call duration and efficiency, while tracking ABANDONED statuses and CALL_TYPE enables a comprehensive overview of call outcomes and trends. Figure 3 represents the call centre overview analysis.

4.2 Station Dashboard

The Call Centre Requests Station Business Intelligence Dashboard provides insights into call performance from the perspective of station devices, utilizing datasets like CALLSTARTDT for call initiation timestamps and ANSWER_SPEED_SECS for response time analysis. With metrics such as TALK_TIME_SECS and WRAP_TIME_SECS, station efficiency can be assessed, while CALL_TYPE classification offers context for call handling strategies and optimization efforts tailored to specific station devices. Figure 4 represents the station analysis.

4.3 Performance Dashboard

The Call Center Requests Performance Business Intelligence Dashboard provides comprehensive insights into call center operations, utilizing datasets like CALLSTARTDT for call initiation tracking and ANSWERED for agent response status. With metrics such as TALK_TIME_SECS and SERVICE_LEVEL, performance can be assessed, while CALL_TYPE classification offers context for call handling strategies and optimization efforts tailored to specific service types. Figure 5 represents the performance dashboard analysis.

4.4 Queue Dashboard

The Call Centre Requests Queue Business Intelligence Dashboard offers insights into call centre efficiency from a queue management perspective, utilizing datasets like QUEUESTARTDT for tracking call entry times and QUEUEENDDT for call exit times. With metrics such as SERVICE_LEVEL and ABANDONED, queue performance can be assessed,

while CALL_TYPE classification provides context for optimizing queue handling strategies and improving service. Figure 6 represents the waiting queue analysis.

4.5 Quality Assurance

The Call Centre Requests Quality Assurance Business Intelligence Dashboard facilitates the evaluation of call centre performance against predefined quality standards, leveraging datasets like ANSWERED for agent responsiveness and TALK_TIME_SECS for conversation duration analysis. With metrics such as SERVICE_LEVEL and ABANDONED, call quality can be assessed, while CALL_TYPE classification offers context for identifying areas of improvement and ensuring consistent service delivery. Figure 7 represents the quality assurance analysis.

4.6 Operational Performance

The Call Centre Requests Operational Efficiency Business Intelligence Dashboard offers insights into call centre performance metrics such as ANSWER_SPEED_SECS for call response times and TALK_TIME_SECS for conversation durations. By analyzing metrics like QUEUEENDDT and QUEUESTARTDT, operational bottlenecks can be identified and addressed, while SERVICE_LEVEL and ABANDONED statuses provide indicators of overall efficiency and customer satisfaction. Figure 8 represents the operational performance analysis.

4.7 Ad Hoc Analysis

The Call Centre Requests Adhoc Analysis Business Intelligence Dashboard enables flexible exploration and analysis of call centre data, utilizing datasets like CALLSTARTDT for temporal analysis and CALLTYPEID for categorization. By leveraging metrics such as ANSWERED and SERVICE_LEVEL, ad-hoc queries can be performed to uncover insights, trends, and anomalies in call centre operations, supporting data-driven decision-making and optimization efforts. Figure 9 represents the ad hoc analysis.

4.8 Key Indicator Analysis

The Call Centre Requests Key Indicator Business Intelligence Dashboard provides a consolidated view of essential metrics like ANSWER_SPEED_SECS and SERVICE_LEVEL, enabling quick assessment of call centre performance. By analyzing data points such as ABANDONED and CALL_TYPE, key trends and areas for improvement can be identified,

facilitating informed decision-making and strategic planning for optimizing call centre operations. Figure 10 represents the key indicator analysis.

Similarly call centre agent, response, and overall agent performance via service level is showing in Figures 11-13.

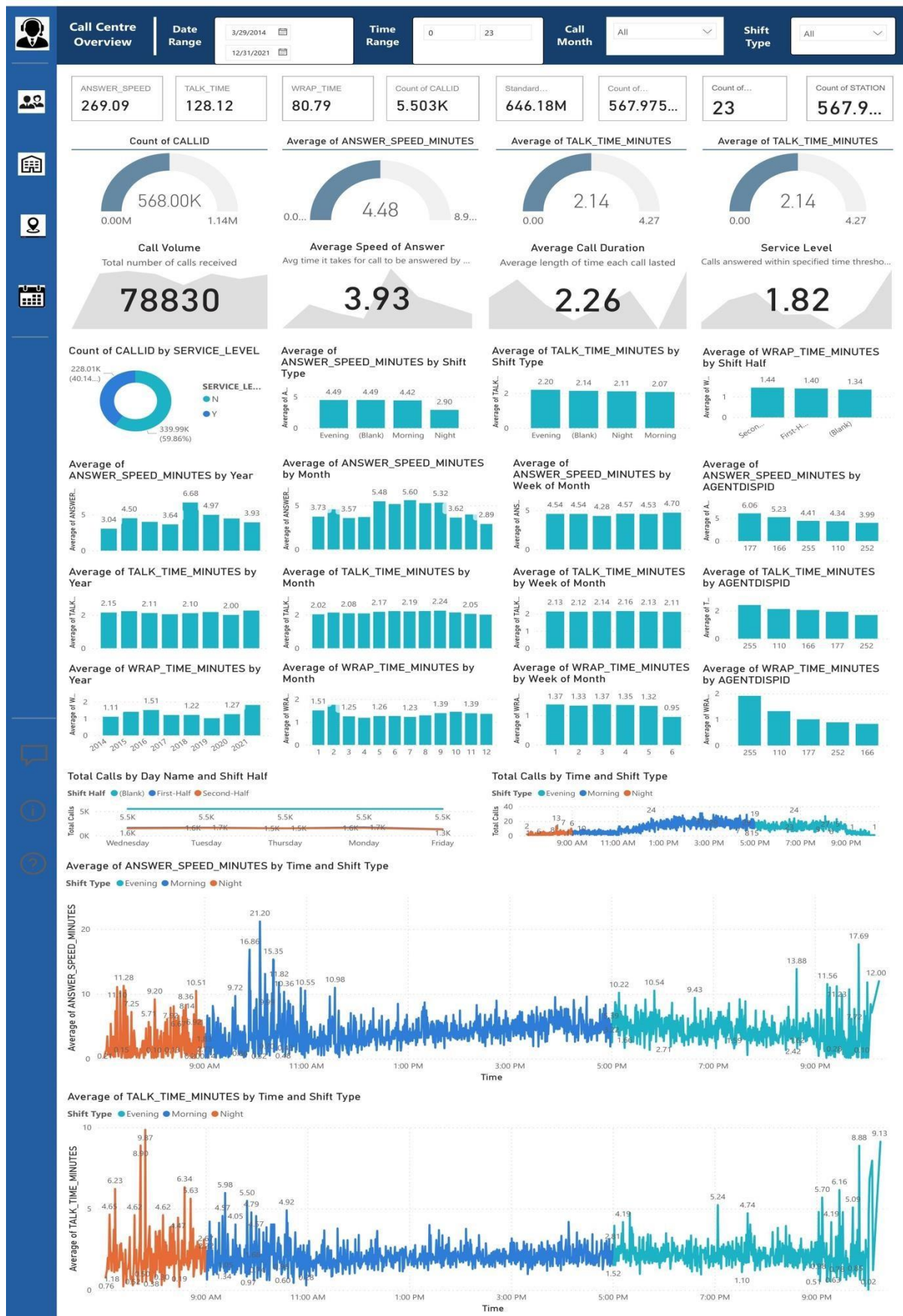


Figure 3 Call centre overview dashboard



Figure 4 Performance analysis of call centre station

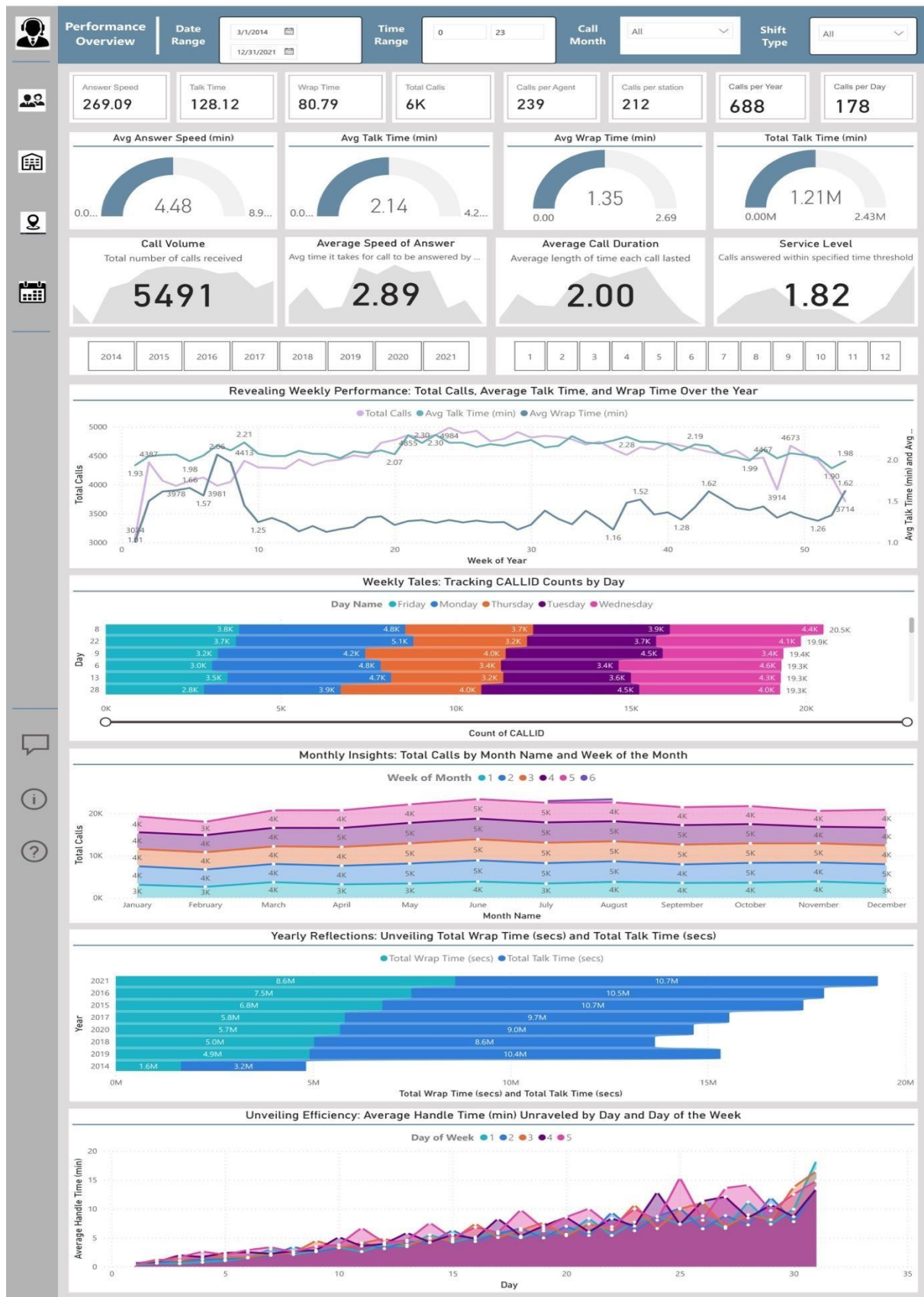


Figure 5 Performance analysis of call centre



Figure 6 Performance analysis of call centre queue process



Figure 7 Performance analysis of call centre quality assurance



Figure 8 Performance analysis of call centre operational efficiency



Figure 9 Performance analysis of call centre ad hoc analysis

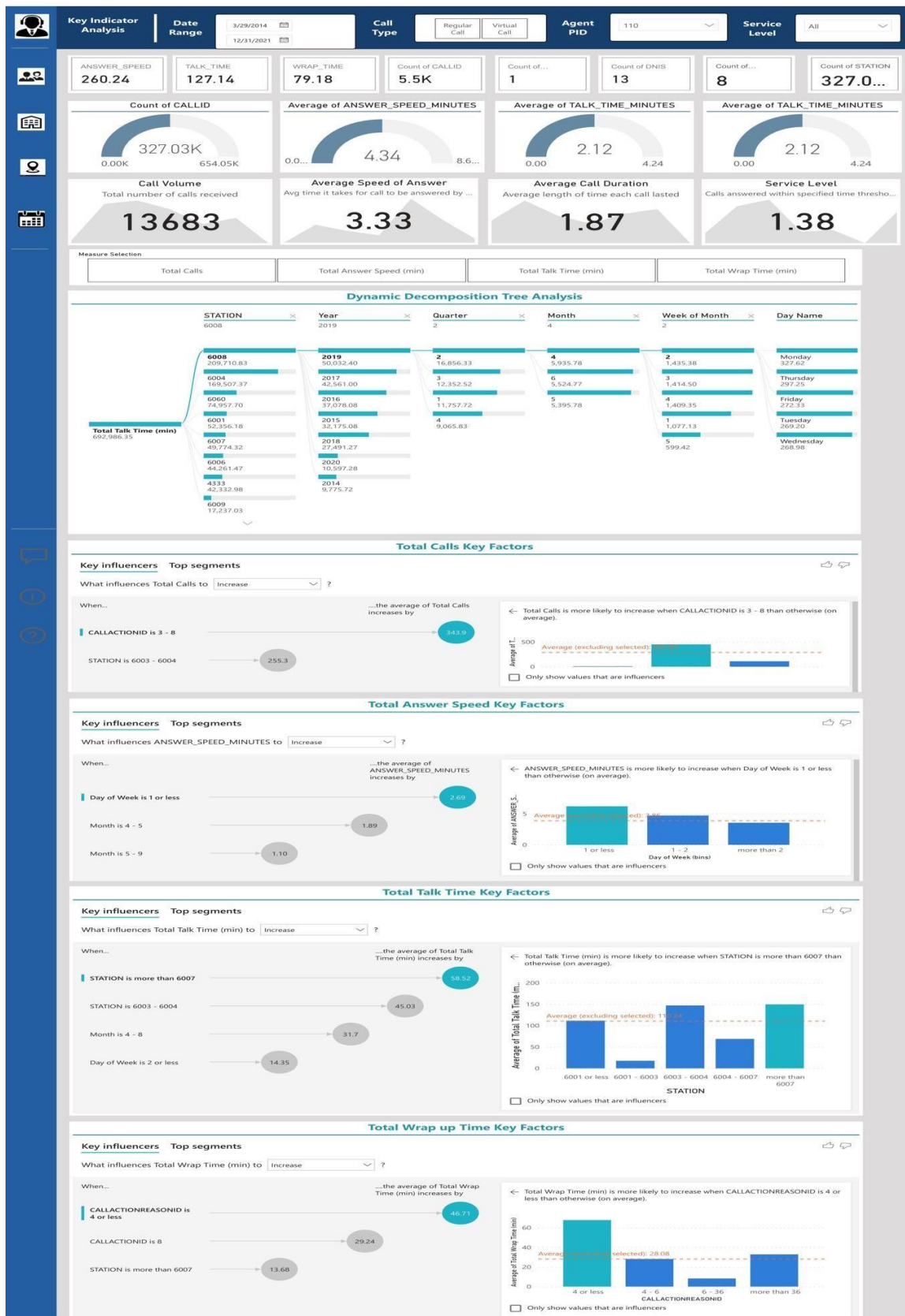


Figure 10 Performance analysis of call centre key indicator analysis

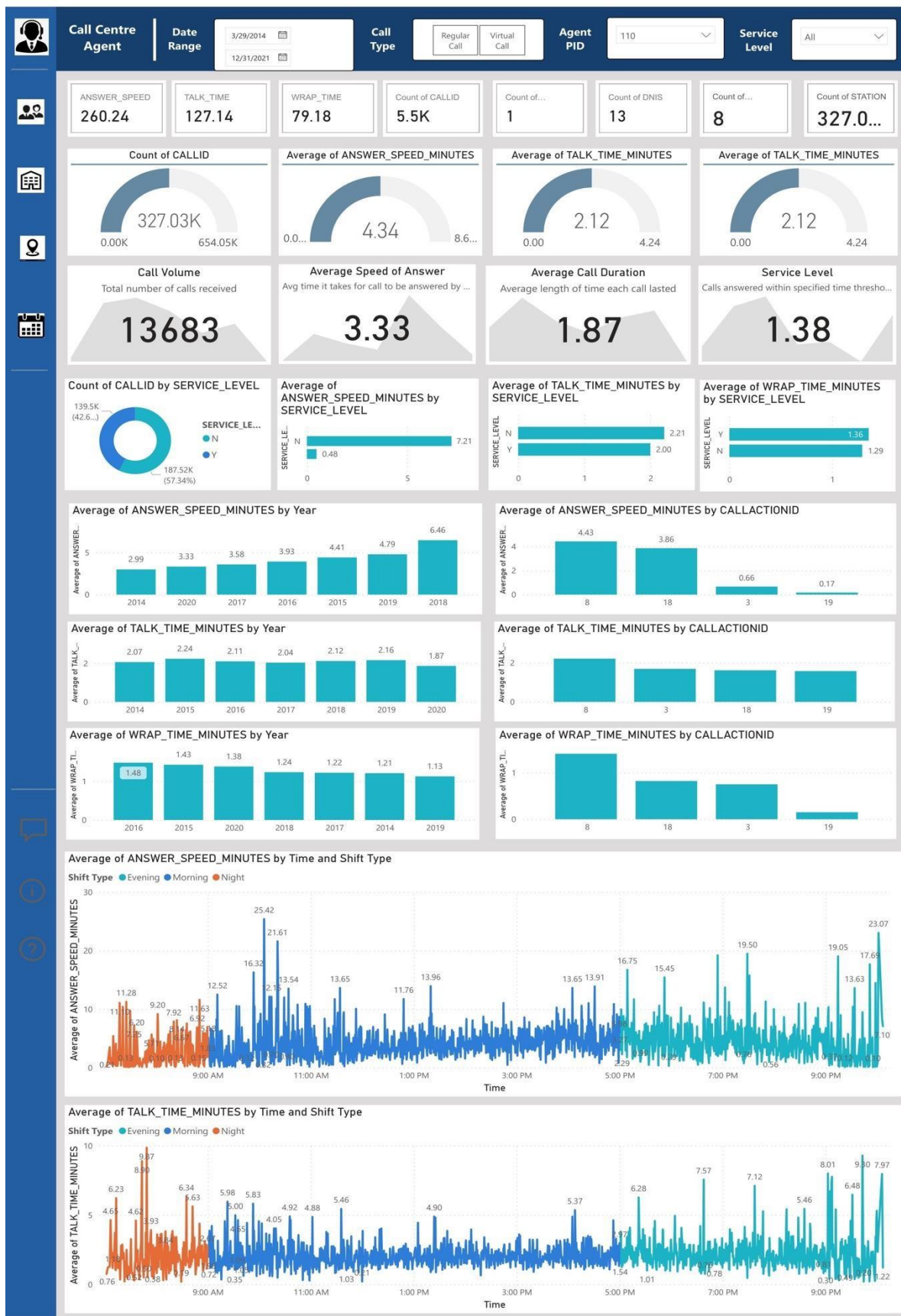


Figure 11 Performance analysis of call centre agent

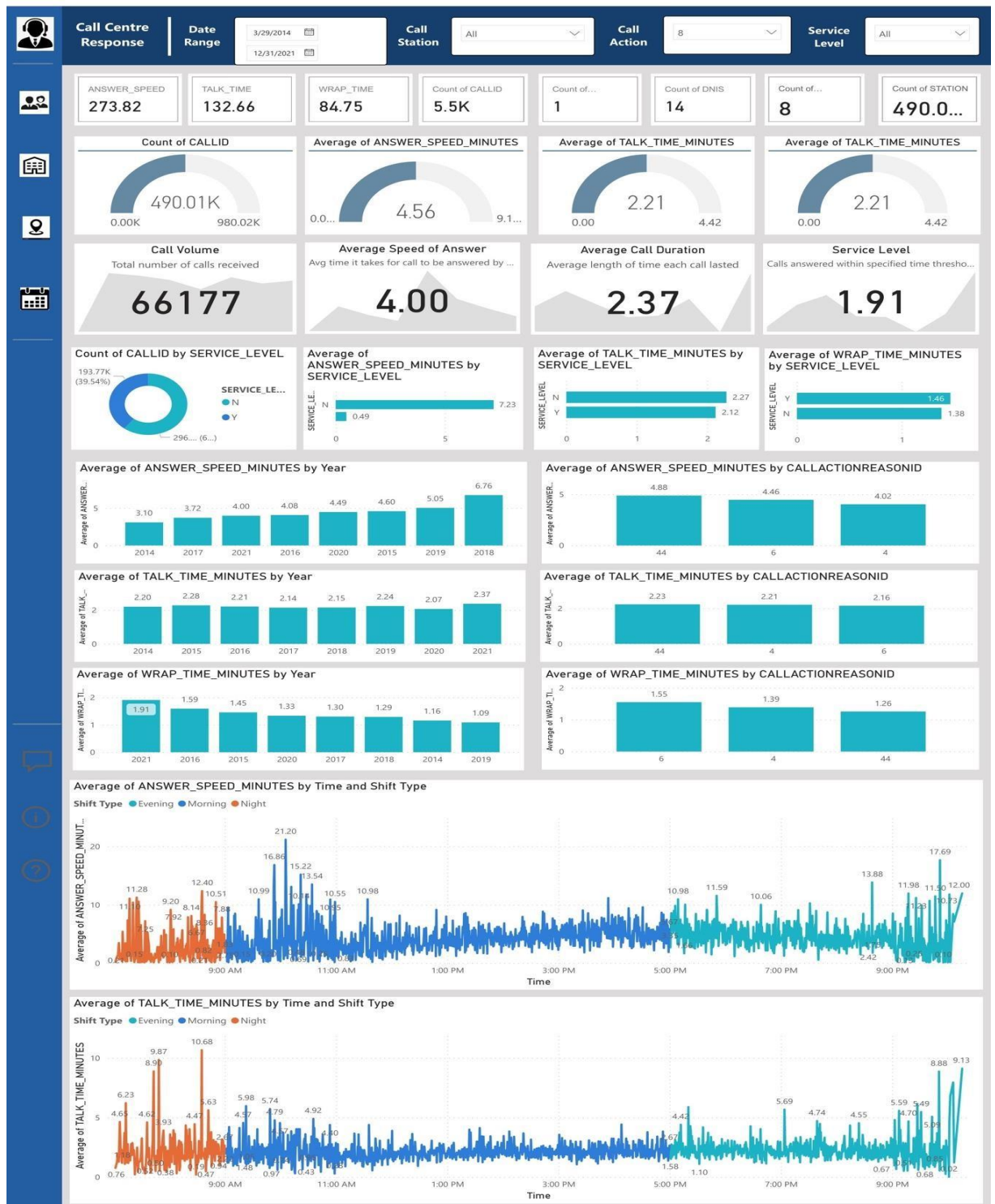


Figure 12 Performance analysis of call centre response



Figure 13 Performance analysis of call centre agent with respect to service level and time

The diagram shows a call centre's entire dashboard, providing a thorough summary of key metrics and data necessary for analysing the efficacy and performance of its operations. This data is useful for analysing the structure in a thesis. Prominent KPIs are highlighted at the top of the dashboard: total calls received, average speed of answer (ASA), average talk time, average handle time (talk time plus post-call work time), average after-call work time, service level percentage (calls answered within a certain amount of time), and total number of abandoned calls. A quick overview of the call centre's general performance and compliance with service requirements is given by these indicators. The graphs provide further detail regarding particular metrics: one shows how talk time varies by service level, another shows how talk time varies within agents, and yet another chart patterns in response time over the course of a week. Additional graphs that provide more information about call patterns and trends are displayed further below. These show the average number of calls that satisfy service level targets per agent, call volume across various time slots and service levels, and total calls received by date and time slot. To help with understanding the complexity of issues mentioned at different times, graphs showing average talk and handling times across different time slots are finally included at the bottom. By providing management with a complete understanding of the call centre's performance, these metrics and visualisations help to identify areas for improvement and speed up decision-making based on accurate operational data.

The image displays an effectively detailed dashboard that has been meticulously developed to gather and analyse a variety of call centre indicators, which makes it a perfect focus for a call reporting thesis. This dashboard is filled with various graphs and statistics that act as visual aids for understanding and analysing the performance data derived from the operations of the call centre. Upon initial inspection, the dashboard displays a top metrics panel that summarises important metrics necessary to assess the effectiveness of the call centre. These comprise the aggregate number of calls received, which offers a broad overview of the volume of calls, and the average speed of answer (ASA), which is essential for assessing the effectiveness of contact handling procedures. Furthermore, the hangup rate which represents the portion of callers who hang up before speaking with an agent is essential for evaluating customer happiness and service availability. Moreover, the average handling time (AHT) and average conversation time metrics provide information about the length of time that agents and consumers spend interacting, providing insight into the effectiveness and efficiency of communication. Additionally, the call center's dedication to timely and effective service delivery is shown by

the service level metric, which shows the percentage of calls answered within a predetermined timeframe.

Several graphs that go deeper into the subtleties of call centre performance can be seen in the dashboard's middle area. A dynamic line chart representing the call volume by date offers a temporal view of call traffic and makes it possible to spot trends and patterns over time. In the meantime, the proportion of calls answered within a predetermined window of time provides a detailed analysis of reaction times and illuminates how quickly the call centre answers incoming calls. Meanwhile, a histogram that displays the distribution of call durations highlights the fluctuations in talk times, which helps understand the complexity of calls and the effectiveness of agents.

As I proceed down into the lower middle part, the focus turns to long-term performance trends that can be seen in year-over-year variations. An analysis of a bar chart that presents the mean response time over a series of years can shed light on how call centre responsiveness has changed over time and help evaluate any operational advancements or setbacks. Analysing average talk times for several years also reveals patterns in customer interactions and may point to changes in call centre operations or consumer behaviour.

Ultimately, comprehensive evaluations of call centre performance across a range of staffing and temporal parameters can be found in the dashboard's bottom portion. Dual-axis line charts allow for the visualisation of long-term trends or seasonal variations by plotting monthly average reaction times and talk times over a period of several years. Furthermore, stratified response and talk times by staff type and time of day assessments provide detailed insights into operational efficiencies and resource utilisation, leading strategic decision-making targeted at improving staff performance and customer service. When taken as a whole, these meticulously developed visualisations provide a thorough overview of important call centre indicators, giving stakeholders the practical knowledge they need to promote operational excellence and continuous development in the call centre setting. In the call centre industry, thorough reporting and analysis play a crucial role in pinpointing performance gaps, developing focused interventions, and eventually raising customer happiness and operational effectiveness.

The image that is being displayed is a large dashboard that comes from a call centre setting. It is carefully crafted to provide a multitude of metrics and information that are necessary for call reporting and analysis. These dashboards are often useful for assessing operational performance, identifying new trends, and developing strategies to improve service quality. This

dashboard's sections and graphs all significantly contribute to illuminating various facets of the call center's operation.

First, the Service Level graph shows the proportion of calls that are answered in a given amount of time. This graph can be used as a gauge for how responsive the call centre is, and the blue coloration may indicate that the service level is being fulfilled. Following that, the Answered/Unanswered Calls bar graph illustrates the difference between calls that are successfully answered and ones that are not, which can be used as a diagnostic tool to see whether there are any staffing deficiencies or call volume mismanagement issues.

subsequently I have Average Answer Speed (Minutes), which provides information about how well call handling procedures are working; a lower average corresponds to quicker response times. The Average Talk Time and Handle Time (Minutes) metrics provide an overview of the average conversation duration as well as the total amount of time spent handling each call, including any follow-up tasks. These kinds of results play a critical role in streamlining operations and improving resource allocation plans.

The Call Volume by Time and Day graph functions as a temporal heatmap, highlighting call influx peaks and troughs at various periods of the day and week. This information helps with capacity planning and workforce scheduling decisions. In the same way, the Average Answer Speed by Time and Item Type graph goes further, breaking down response times according to temporal variables as well as the type of call or item. This allows for a more comprehensive view of service-level performance in a variety of scenarios.

The Average Talk Time by Year graph, which shifts the focus to longer-term trends, provides a historical perspective on call duration averages and may reveal tendencies of regression or improvement over time. On the other hand, the Average Talk Time by Call/Item Type graph identifies areas that require concentrated attention or optimisation efforts by breaking down average talk times by certain call or item categories.

The Average Talk Time by Time and Shift Type graph breaks down talk time averages into distinct shifts and times, offering crucial information about the effectiveness of different staffing configurations and the possible necessity of focused training interventions. When taken as a whole, this dashboard proves to be a vital tool for contact centre management, giving stakeholders the knowledge and skills they need to improve operational plans, guarantee quality control, and maintain the highest levels of customer service.

The image exhibited a detailed dashboard that comes from a call centre setting. It is carefully designed to contain a multitude of metrics and statistics that are intended to shed light on the

call center's performance. Every graph and collection of metrics that is carefully arranged in the dashboard has a specific function, be it evaluating performance trends over time, comparing performance in various categories, explaining distribution patterns, or summarising other relevant metrics related to call centre operations.

A row of classic KPIs sits at the top of the dashboard and is vital for determining how well the call centre is doing in terms of efficiency and customer focus. A critical benchmark for managerial decision-making and strategic planning, metrics such as calls offered, calls handled, average speed of answer (ASA), service level, call abandon rate, and average handle time (AHT) provide an overview of the center's operational effectiveness and its level of customer interactions.

A series of graphs that provide detailed insights into various aspects of call centre performance unfold beneath the broad measures. The graphs labelled Count of ANSWER_SPEED_MINUTES by Month and Count of ANSWER_SPEED_MINUTES by Year provide a comprehensive overview of how response times vary on a yearly and monthly basis by carefully analysing the distribution of call answering speeds along various temporal dimensions. The Average of Talk Minutes by Year graph provides a longitudinal view of average talk times annually to complement these temporal analyses. The Average of Talk Minutes by CALL Reason Code and Average of Talk Minutes by Service Type graphs, on the other hand, break down talk time averages by call reason code and service type, respectively, explaining the relative time commitment of various call categories and service offerings.

Time series analyses are used to create graphs like the Average of ANSWER_SPEED_MINUTES by Time and Shift Type and Average of TALK_MINUTES by Time and Shift Type, which capture the daily ups and downs of call centre operations by showing differences in call durations and answering speeds between different time slots and shift types. These understandings of the temporal subtleties of contact centre operations are crucial for improving shift planning techniques, allocating resources optimally, and guaranteeing the flawless provision of exceptional customer support all day long.

All of the dashboard's visualisations work together to extract valuable information that gives call centre managers the knowledge they need to improve customer satisfaction, increase efficiency, and strengthen performance. It is essential to note that these graphs' interpretive value depends on how they are compared to past data, predetermined performance thresholds, or the particular goals and aspirations of the call centre. This highlights the natural alignment

that exists between data-driven decision-making and operational excellence in the context of call centre management.

The image depicts a complete dashboard with an abundance of graphs and metrics. It is clear that this dashboard was created with the careful observation and comprehensive reporting of contact centre or telecom operations in mind. A dashboard of the sort is an essential tool for providing managers or analysts with a quick summary of key performance indicators (KPIs), which helps with strategic planning and decision-making. Examining every aspect of this complex dashboard reveals an extensive number of information.

A series of Key Performance Indicators (KPIs) are positioned prominently on the dashboard; each KPI acts as a barometer to measure the operational efficiency and customer-centricity of the call centre. Key dimensions like call volume, responsiveness, service quality, and forecasting accuracy are captured by metrics like Total Calls, Average Answer Speed, Service Level, Average Handle Time, Abandon Rate, Call Forecast, and Forecast Accuracy, which provide stakeholders with a comprehensive picture of call centre performance.

Above the broad KPIs, there is a plethora of graphs and indicators that are carefully designed to provide subtle insights into various aspects of call centre operations. The Call Volume vs. Answered by Day of Week chart allows for the discovery of temporal trends that require staffing modifications by contrasting the overall volume of calls with the number of calls that are effectively answered on various days of the week. In the same way, the average call/wait time and average wrap time bar graphs provide a detailed breakdown of average call/wait times and post-call wrap-up times, which are vital for streamlining service delivery procedures and enhancing client interactions.

Delving deeper into the web of data visualisation, the Total Calls/Wait Time by Month and Day line chart reveals monthly patterns in call volume and corresponding wait times, and the Total Answer Speed By Group bar chart breaks down average response times into different departments or groups within the call centre environment, facilitating performance benchmarking within the organisation. In contrast, the Total Call Distribution By Hour heatmap functions as a temporal heatmap, showing the peaks and valleys of call volume throughout the day, which is crucial for managing personnel levels and maximising operational effectiveness. delve further into the combination of data visualisation: the Average Wait Time by Month bar graph and Average Wait Time By Queue pie chart reveal the distribution of average wait times over different temporal epochs and call queues, respectively, while the Total Answer Speed By Day Of Week graph provides a temporal breakdown of average answer speeds over different

days. In conclusion, the calendar view titled "Total Call Volume by Day" provides an overview of daily call levels and makes it easier to spot unusual peaks or troughs in call volume, which is crucial for proactive resource allocation and operational flexibility. To give stakeholders a complete picture of call centre performance, this painstakingly constructed dashboard essentially combines historical and real-time data. It is a vital tool for strategic decision-making, performance optimisation, and resource planning, and it represents the mutually beneficial interaction between operational excellence and data-driven insights in the ever-changing field of call centre management.

The image provides a detailed understanding of a dashboard that is clearly intended for call reporting. It might have been customised to enable analysis in the communications, customer service, or similar industries. Including a variety of indicators and graphical displays, the dashboard appears to address call statistics, most likely meeting the needs of customer service departments or call centres. Every component contained in the dashboard provides a detailed understanding of many aspects of call centre operations, forming an intricate web of data-driven explanations. The dashboard's topmost section features a row of block-style metrics that outline critical elements of call centre performance. These metrics start with Total Calls and Average Answer Speed and go down to Average Talk Time and Average Handle Time, and finally, Outbound Calls and Calls In Queue. The concisely arranged data provide managers and analysts with a rapid snapshot of call centre efficiency and operational health, acting as a compass. Delving further into data visualisation, an abundance of graphs and charts provide a rainbow of information about call type distributions, performance swings, and temporal trends. With an excellent objective of reducing client wait times, the Total Answer Speed by Month (Year To Date) bar graph, for example, attempts to monitor the call center's response over the course of several months. On the other hand, the MTD (Month To Date) Total Answer Speed by Day line graph aims to decipher the daily fluctuations in responsiveness, maybe revealing temporal patterns or aberrations that require additional study.

Meanwhile, the line graphs representing Total Talk Time and Total Work Time (Year To Date) and (Month To Date) show how call centre workload and productivity have changed over long and short time periods, providing information about the seasonal patterns in engagement and productivity. At the bottom of the dashboard, pie charts and bar graphs explain the volumetric distribution of different call categories, the breakdown of call types, and the diurnal subtleties of answer speed distribution. These data are essential to guiding resource allocation strategies and service level optimisation programmes.

There are probably more elements and interactive features added to these visual representations with the intention of improving user experience and analytical depth. The dashboard's usefulness can be increased by adding filters, drill-down capabilities, and comparing features. These features allow users to identify more precise patterns, gain deeper insights, and extract relevant information for call centre performance optimisation efforts. The dashboard proves to be a true cure-all for call centre management, providing stakeholders with a wide range of call centre performance data, from workload analysis and call type distribution to call volume and responsiveness. The secret to operational effectiveness and customer-centric excellence can be found embedded in these visual and metric subtleties, which support the fundamental requirement of data-driven decision-making in the dynamic pressure cooker that is call centre operations.

The image shows a dashboard created to give a thorough overview of analytics and performance indicators related to call centres. The dashboard is divided into multiple parts, each of which provides distinct insights into various facets of the call center's activities. Key performance indicators (KPIs) that provide a rapid snapshot of operational efficiency and customer contact dynamics are displayed at the top. These KPIs include total calls, average time, abandoned rate, and average speed of answer. A huge number of graphs and charts, ranging from hourly variations in incoming versus outgoing calls to weekly performance tracking and monthly insights, are displayed beneath these KPIs to reveal temporal trends, performance changes, and operational intricacies. The dashboard gives call centre managers the tools they need to track, evaluate, and improve operational effectiveness using a variety of visual aids. This promotes well-informed decision-making and raises the standard of customer support.

Strategic planning and managerial oversight benefit greatly from the dashboard's comprehensive breakdown of call centre performance metrics and data. The dashboard's graphs and metrics offer insightful information on call volume, responsiveness, workload distribution, and efficiency across a range of temporal dimensions. Call centre managers can see trends, deal with problems, project staffing requirements, and ultimately enhance the efficiency and calibre of customer care by using this thorough understanding of operational dynamics. The dashboard thus becomes a vital tool in the call centre environment's pursuit of operational excellence. The picture depicts a well-designed dashboard that includes several metrics and data visualisations related to a call center's performance assessment. It was most likely created to support a call reporting thesis or analytical investigation. Every panel on the dashboard

reveals a different component of the operating dynamics of the call centre and provides in-depth information about productivity, effectiveness, and service target adherence. The dashboard's highest point is the Top Metrics Panel, which contains key performance indicators (KPIs) like "Call Volume" and "Average Speed of Answer." These KPIs provide stakeholders with a brief summary of operational performance metrics that are essential for strategic decision-making and managerial oversight.

The dashboard presents a variety of visualisations, each designed to explain a particular aspect of call centre operations. These visual aids for workload analysis, resource management, and performance evaluation range from the Agent Status Time Breakdown Bar Chart, which breaks down the time that agents spend on different statuses, to the Service Level vs. Abandoned Calls and Service Events Line Chart, which explores the complex relationship between service quality and customer satisfaction. Additionally, the After Call Work (ACW) Stacked Bar Chart and Task Wrap Time explain the post-call activities of agents, and the Handle Time Trends with Quarterly Insights Line Chart identifies seasonal differences in call handle times to aid in forecasting and operational optimisation initiatives.

All of these illustrations work together to make complex data sets easier to understand and use for those who have an interest in the call center's performance. The Dynamic By Hour By Day Heatmap facilitates workforce planning, while the Monthly Performance vs. Target bar chart displays performance benchmarking. All of these graphs provide insight into important aspects of call centre performance, enabling stakeholders to make well-informed decisions and verify theories put forth in the field of call reporting. In final analysis, the dashboard exemplifies a persuasive combination of analytical skill and data-driven insights, highlighting its critical role in bolstering operational effectiveness and service quality in the call centre environment.

The dashboard provides an extensive range of metrics and graphs intended to analyse and assess call reporting patterns; these could be used in a thesis or performance study on customer service or call centre effectiveness. Divided into discrete sections, every element of the dashboard provides stakeholders with a detailed comprehension of the key performance indicators (KPIs) that are essential for evaluating the efficacy of call handling and operational efficiency. Crucial data including the total number of calls, the answered and abandoned rates, and performance levels are displayed in the top metrics overview, giving an overview of call centre operations.

The performance summary graphics then go into further detail on a number of KPIs, such as abandonment rate, average speed of answer (ASA), answer seizure ratio (ASR), and service

level. Stakeholders can see patterns and trends in call efficiency and responsiveness using these visualisations, which enables focused interventions and operational modifications. Additionally, the dashboard is divided into parts for total calls, answered calls, abandoned calls, and average call duration. Each of these sections provides a detailed analysis of relevant data along with corresponding histograms that clarify temporal distributions. These observations are crucial for managers in terms of staffing levels, resource allocation, and pinpointing areas in the call centre environment that need improvement. In summary, the dashboard functions as a potent analytical tool that gives stakeholders the ability to thoroughly examine call centre operations, verify theories, and make data-driven choices. This dashboard supports operational excellence and enhances customer service delivery by offering a comprehensive perspective of call handling dynamics. This strengthens the validity and resilience of results found in call reporting studies or performance evaluations.

The image consists of snapshots of a call reporting data dashboard, taken from a business intelligence interface used to monitor performance indicators in a call centre or inside a particular organisation. The five main metric indicators that are most important for evaluating operational effectiveness and customer service quality are call volume, duration, talk time, wrap time, and hold time. Every graph on the dashboard provides a thorough study of the dynamics of the call centre over time, making trend analysis and performance assessment easier.

The first graph provides a long-term perspective on call volume patterns by showing the total number of calls received and recorded by month. The second graph that follows shows the average length of calls per month and sheds light on the intricacy of the topics covered or the effectiveness of call handling. In the meanwhile, the following graphs provide information on staffing requirements, post-call work responsibilities, and client wait times by delving into the total talk time, wrap time, and hold time by month. Interactive components like tooltips are added to these visual representations to improve the user's capacity to perform in-depth analysis and extract useful insights from the data. The graph provides a thorough summary of call centre performance indicators, highlighting monthly variations and temporal patterns. Managers and analysts can use it as a useful tool to comprehend operational dynamics, spot inefficiencies, and plan changes that would boost customer satisfaction and service performance in call centres. This dashboard is probably a component of a larger research or thesis that examines call handling practices and how they affect organisational efficacy in the context of customer service.

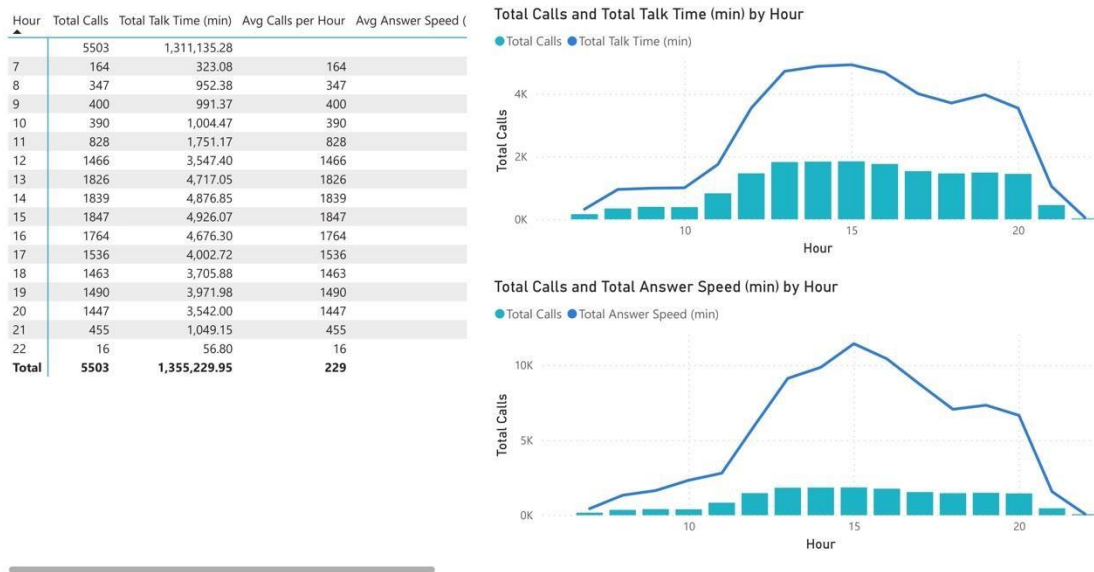


Figure 14 Performance analysis of call centre station

Figure 14 provides several call reporting data. It consists of a table with three corresponding graphs that provide information on call-related metrics over a given period of time. Prominently shown on the left side of the picture, the table displays raw data in six columns: Hour, Total Calls, Total Talk Time (min), Average Calls Per Hour, Average Talk Time/Call, and Average Answer Speed. With indicators like contact volume, conversation duration, and average response speeds included, this tabular data provides a thorough picture of call centre activities and serves as a foundation for further analysis and assessment. Three graphs that clearly show important components of call centre performance support the table. The first graph shows the total calls and the total talk time (min) by hour using a combination of bar and line elements. This graph makes it easier to comprehend how call centre activity patterns change over time by providing insights into the relationship between call volume and the length of talks throughout operating hours. The second graph, which shows Total Calls per hour as a simple bar graph, gives a clear picture of the amount of calls without any more context. The third graph, which shows the total calls and total answer speed (min) by hour, is a combination line graph.

Figure 15 helps analysts allocate resources and evaluate performance in the call centre environment by showing how service levels change in response to different call volumes by establishing a correlation between call volume and answer speed. When combined, these visual aids provide analysts with insightful information that helps them recognise patterns, identify

peak times, and assess the general performance of call centres. This information supports wellinformed decision-making and operational optimisation.

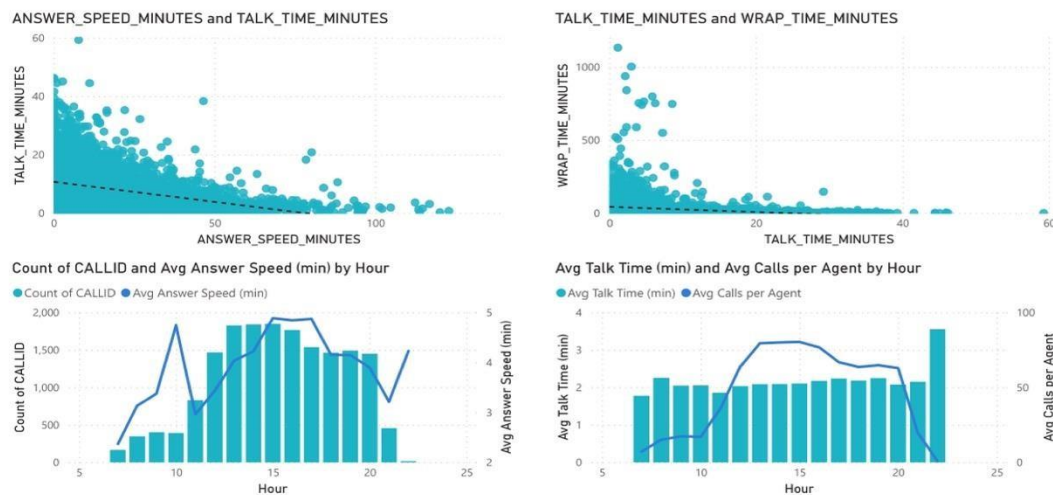


Figure 15 Performance analysis of call centre station

Four different graphs are displayed in the image, each providing insights into call reporting data. These graphs were probably created for analytical or thesis-related objectives. The first scatter plot shows the link between Talk Time (Minutes) and Answer Speed (Minutes), showing a negative trend that suggests calls that are answered more quickly typically last less time. A more complex link between the two variables is shown in the second scatter plot, which investigates the correlation between Talk Time and Wrap Time (Minutes). The third graph, which shows the Average Answer Speed and the Count of Call IDs by Hour together with variations in call volume and average response times throughout the day, is a hybrid column chart and line graph. The average talk time per call and the average calls per agent by hour are displayed in the fourth graph, which is another combination figure that provides information about staffing requirements and task allocation over time.

Collectively, these graphs offer a comprehensive review of call centre dynamics, including agent task distribution, response times, and call handling durations. These kinds of information are critical to improving customer satisfaction, managing human resources in a call centre setting, and optimising call centre performance. Organisations may enhance operational efficiency and service quality and ultimately improve customer and call centre staff outcomes by using these critical KPIs to guide their decision-making process.

Chapter 5

Conclusion

5. Conclusion

This report, with its particular focus on call centre reporting system optimisation, represents an important addition to the field of agile techniques in business intelligence and data warehousing. Using a comprehensive analysis and combination of an extensive range of academic research concerning agile software development, data modelling, and business intelligence methodologies, this study has highlighted the innovative ability of agile concepts to augment operational effectiveness, cultivate stakeholder cooperation, and ultimately guarantee project success. Furthermore, the comparison between agile approaches and conventional waterfall methodologies has focused on the particular advantages that agile practices can offer business intelligence initiatives in addition to demonstrating the inherent benefits of agile methodologies in IT projects. This thesis has successfully illustrated the concrete influence that agile approaches may have on enhancing data-driven decision-making processes, particularly in the context of call centre operations, by providing real-world examples and case studies from the literature. In addition, the study of new ideas like automated multidimensional modelling, data vault methodology, and semantic approaches to schema design has expanded the scope of conventional data warehousing techniques, providing innovative solutions to challenging data modelling problems and advancing the field of data modelling.

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